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Electric Vehicle
Codec Technologies India
Internship 2026

EV Drivetrain Modelling and Simulation

Battery -> inverter -> PMSM / IM / brushed DC -> gearbox -> wheel,
backward and forward simulation
in Python

AT A GLANCE

95.0 / 92.8 / 91.0 %

Rated efficiency PMSM / IM / DC

142.6 / 146.1 / 166.4

WLTP-like Wh/km PMSM / IM / DC

117.6 / 127.5 / 153.1

MIDC-like Wh/km PMSM / IM / DC

253 / 246 / 216 km

Range WLTP-like, 36 kWh usable

10.3 / 10.2 / 12.4 s

0-100 km/h PMSM / IM / DC

31 % of traction

Regen recovered, WLTP-like

< 1e-9 Wh

Energy-balance residual

15 / 15

Tests re-run here

VERDICT

TESTS 15/15

RE-RUN HERE

DECLARED SUBSTITUTION

SYNTHETIC CYCLES

NO SIMULINK

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1. Objective and brief mapping

Brief: simulate the entire drivetrain including motor, inverter and wheels using MATLAB/Simulink or Ansys; model motor torque, gear ratios and energy losses; compare AC vs DC motor efficiencies. I did not have a MATLAB/Simulink licence, so I built the Simulink model as an **equivalent block-by-block Python package** (drivetrain/): vehicle longitudinal dynamics, single-speed gearbox, three quasi-static ~80 kW machine models (PMSM, squirrel-cage induction, separately-excited brushed DC), a 3-phase VSI loss model (IGBT and SiC) plus a 2-quadrant chopper for the DC motor, an Rint 40 kWh battery and a PI driver. The AC-vs-DC question is answered with numbers over two drive cycles. python main.py regenerates results/ in about 13 s.

2. Model and architecture

Vehicle: 1,580 kg (Leaf-class), C_{dA} 0.636 m², C_{rr} 0.010, r_{wheel} 0.316 m, mass factor 1.05, traction limit μ 0.9 x 55 % axle share. Gearbox η 0.97, ratio chosen per motor so that ω_{max} maps to 150 km/h: 8.25 (PMSM, Leaf uses 8.19), 9.53 (IM), 5.16 (DC). Losses: $P_{fe} = (k_h w + k_e w^2)(\phi/\phi_0)^2$, $P_{mec} = k_f w + k_w w^3$, stray $P_{s0} (T/T_{max})^2$, plus copper (PMSM $1.5 R I^2$ with field-weakening I_d), rotor slip loss $s/(1-s) P_{mech}$ (IM), armature $I^2 R + 2 V$ brush drop + 1.5 kW field (DC). Inverter per-switch conduction + switching losses from Infineon AN2008-03 with FS400R07A3E3 (IGBT, 10 kHz) or CAB450M12XM3 (SiC, 20 kHz) data.

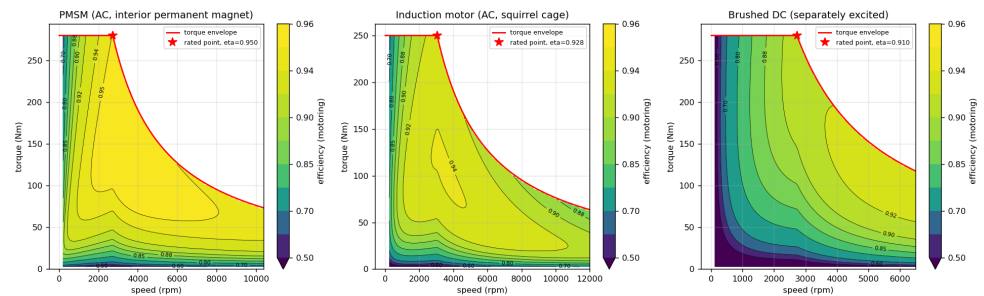


Fig. 1 - Analytical efficiency maps of the three machines with torque-speed envelopes and rated points (95.0 / 92.8 / 91.0 %).

3. Method and implementation

- Backward (quasi-static) path: $F_{req} = m_{eff} dv/dt + F_{roll} + F_{aero} + F_{grade}$ from the cycle, $T_{motor} = T_{wheel}/(g \eta_{gb})$, losses summed per sample; energy bookkeeping $P_{batt_chem} = P_{wheel} + P_{gearbox} + P_{motor} + P_{inverter} + P_{aux} + P_{battery}$ checked every step.
- Forward path: PI driver $F_{cmd} = K_p e + K_i \text{integral}(e) + \text{feed-forward}$ (K_p 800 N per m/s, K_i 80 N/m, anti-windup), command split into motor torque within the envelope and friction braking, 0.1 s fixed step.
- Drive cycles are **synthetic** piecewise-linear micro-trip sequences tuned to the phase durations, maximum speeds and distances of WLTC class 3b (1,800 s, 23.49 km vs 23.27 real, v_{max} 131.3 km/h) and the Indian MIDC (1,180 s, 10.33 km vs 10.65 real, 90 km/h); any Wh/km is therefore WLTP-like, not a homologation value.
- Battery: Rint 96s2p, 40 kWh, 80 mOhm, 4-point OCV table, 90 % usable window, 300 W auxiliaries; range = 36 kWh / (Wh/km).

4. Results (results/summary.json, comparison_table.csv)

PMSM energy flow on the WLTP-like cycle (Wh over 23.49 km): battery chemical 3,348, battery $I^2 R$ 63, inverter 214, motor 450 (copper 112, iron 279, mechanical 57, stray 2), gearbox 137, auxiliaries 150, wheel traction 3,402, wheel regen recovered 1,068 (31.4 % of traction), friction brakes 6.8 Wh; balance residual 1.4e-12 Wh, 0 envelope violations. Gear-ratio sweep (PMSM): 0-100 km/h 14.9 s at 4:1, 10.3 s at 8.25, 10.1 s at 9-12 (tyre-limited at 7.7 kN); top speed 204 km/h road-load limited below 6:1, 150 at 8.25, 103 at 12. SiC inverter cuts inverter loss 214 -> 99 Wh and consumption 142.6 -> 137.6 Wh/km (-3.5 %).

Metric	PMSM (AC)	Induction (AC)	Brushed DC
Gear ratio for 150 km/h	8.25	9.53	5.16
Rated-point efficiency (80 kW)	95.0 %	92.8 %	91.0 %
Rated losses Cu / Fe / mech / stray / other (W)	2,880 / 1,000 / 88 / 200 / 0	2,754 / 1,030 / 99 / 300 / 2,051 rotor	3,675 / 1,408 / 289 / 300 / 2,200 brush+field
Cycle motor efficiency WLTP-like / MIDC-like	90.4 / 86.5 %	89.8 / 85.4 %	80.1 / 70.1 %
Battery-to-wheel efficiency WLTP-like	80.4 %	79.4 %	73.2 %
Consumption WLTP-like / MIDC-like	142.6 / 117.6 Wh/km	146.1 / 127.5 Wh/km	166.4 / 153.1 Wh/km
Range WLTP-like / MIDC-like	253 / 306 km	246 / 282 km	216 / 235 km
0-100 km/h / top speed	10.3 s / 150 km/h	10.2 s / 150 km/h	12.4 s / 150 km/h
Inverter loss WLTP-like	214 Wh (6-switch VSI)	240 Wh	88 Wh (chopper)
Forward vs backward Wh/km, tracking RMSE	142.3 vs 142.6, 0.03 km/h	145.8 vs 146.1	166.2 vs 166.4

KEY TAKEAWAY With every motor geared for the same 150 km/h, the brushed DC machine gives up 17-30 % range to the PMSM and the induction motor 2-8 %; the Python model replaces Simulink block-for-block, declared in the report.

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PMSM gear ratio	4	6	8.25 (design)	9	10	12	12.5+
0-100 km/h (s)	14.9	11.5	10.3	10.1	10.1	10.1	unreachable (motor over-speed)
Top speed (km/h)	204 (road-load limited)	204	150	138	124	103	< 100

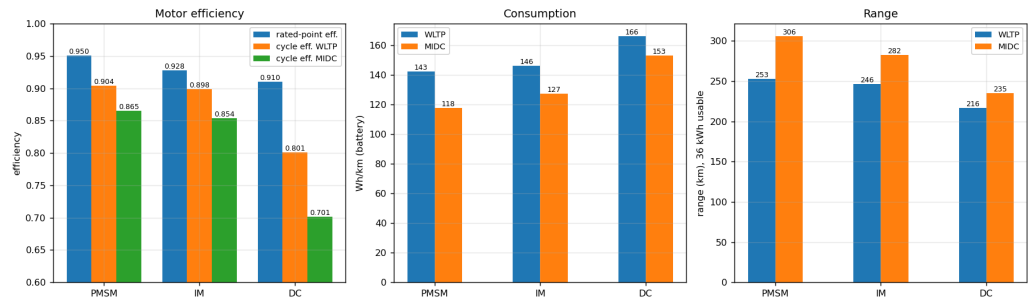


Fig. 2 - AC vs DC comparison: cycle efficiency, Wh/km and range for the three machines on both cycles.

5. Verification

Re-run here with the fixture-aware runner (pytest not installed): **15 passed, 0 failed** in 1.0 s.

Test	Checks	Result
test_energy_conservation_per_sample, test_forward_simulation_conserves_energy, test_motor_electrical_power_is_mech_plus_losses	$P_{batt_chem} = P_{wheel} + losses + P_{aux}$ at every sample, backward and forward	3 PASS
test_losses_non_negative_and_efficiency_in_unit_interval, test_cycle_efficiencies_in_unit_interval, test_inverter_efficiency_range	every loss ≥ 0 ; map, rated, cycle and inverter efficiencies in (0, 1]	3 PASS
test_pmsm_more_efficient_than_brushed_dc, test_pmsm_at_least_as_efficient_as_induction	AC vs DC ordering on cycle efficiency and Wh/km	2 PASS
test_higher_gear_ratio_faster_0_100_but_lower_top_speed, test_top_speed_limited_by_motor_speed_at_design_ratio	5:1 vs 8:1 launch, 8:1 vs 12:1 top speed, 150 km/h at design ratio	2 PASS
test_tractive_force_at_least_resistive_at_steady_state, test_no_envelope_violations_on_cycles	motor force $\geq$ road load at cruise; 0 envelope violations on both cycles	2 PASS
test_rint_battery_current_reproduces_power, test_torque_envelope, test_gearbox_loss_sign_aware	I from P exact; $T \leq \min(T_{max}, P_{max}/w)$; traction vs regen loss sign	3 PASS

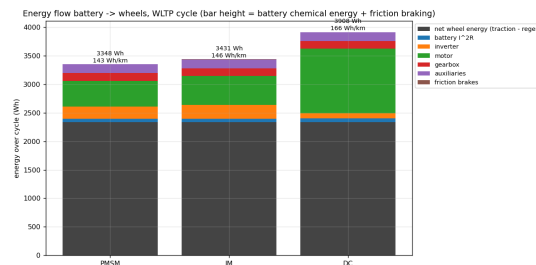


Fig. 3 - Energy flow on the WLTP-like cycle for the three motors (Wh): battery chemical energy split into wheel work, regen recovered and each loss term.

6. Data integrity and declared substitutions

■ **MATLAB/Simulink and Ansys not used** - README: "I did not have a MATLAB/Simulink licence for this internship, so I built the Simulink model as an equivalent block-by-block Python package." Section 8 of the README maps each module to its Powertrain Blockset / Simscape block (Drive Cycle Source, Longitudinal Driver, Vehicle Body 1DOF, Simple Gear, Mapped Motor / PMSM / Induction Machine / DC Machine, Average-Value Inverter, Datasheet Battery). Ansys Motor-CAD / Maxwell would replace motors.py with an FE loss map; nothing here ran in either tool.

■ Drive cycles are synthetic approximations of WLTC 3b and MIDC, not the official second-by-second tables. Motor loss coefficients are my assumptions tuned to published rated-point efficiencies (Leaf EM57 rating for the PMSM; IM and DC assumed same class); quasi-static models with no dq dynamics, saturation, temperature or PWM ripple; Rint battery without thermal or ageing effects.

7. Learnings

The brushed DC machine loses mainly through constant field excitation, brush drop and brush friction that do not scale down with load - exactly why DC traction motors disappeared from production EVs, and the MIDC-like (low-load) cycle punishes it hardest (70 % cycle efficiency). The induction motor stays within 2-8 % of the PMSM; its penalty is the always-present magnetising current and rotor slip loss. The DC chopper is cheaper in losses than the VSI, but an order of magnitude less than the motor penalty. Above about 9:1 the launch is tyre-limited, so a higher ratio only costs top speed. Forward and backward formulations agree within 0.2 %, which is what I expected when the driver never saturates.

KEY TAKEAWAY With every motor geared for the same 150 km/h, the brushed DC machine gives up 17-30 % range to the PMSM and the induction motor 2-8 %; the Python model replaces Simulink block-for-block, declared in the report.